

Teaching Manual: Graphs and Data Displays

Scripted lesson | 60-75 minutes

LESSON OVERVIEW

Learning goal: Collect, display, and interpret data using scaled graphs, line plots, and simple circle graphs.

Success Criteria

- The student represents the idea with a model, drawing, table, or number line.
- The student uses the lesson vocabulary accurately.
- The student explains why a strategy works and checks reasonableness.
- The student can respond independently to: Why must you read the key before answering questions about a scaled graph?

Teaching Theory

Measurement compares an attribute with a chosen unit. Data displays encode quantities through scales, keys, positions, and categories, so units and labels are part of the mathematics.

Instructional principle: Move concrete -> pictorial -> abstract. Do not remove the model until the student can explain the symbolic work.

Lesson Flow

5-8 min	Number talk
10 min	Vocabulary and worked example
20 min	Main lesson
15 min	Practice
10 min	Challenge
5 min	Exit check

Materials: Preparation and Use

BEFORE THE STUDENT ARRIVES

Survey question	Prepare before the lesson and introduce it only when it supports the current mathematical idea.
graph paper	Place beside the student. Use for drawings, organized records, and a second representation.
colored pencils	Prepare before the lesson and introduce it only when it supports the current mathematical idea.
weekly assessment.	Prepare before the lesson and introduce it only when it supports the current mathematical idea.

Table Setup

- Center: the main model or tool for today's lesson.
- Student side: packet, pencil, and only the materials currently needed.
- Teacher side: answer notes and observation record, kept out of student view.
- Leave a clear space where the student can point, move pieces, and explain.

Vocabulary to Establish

unit: Ask the student for an example or picture before refining the definition.

estimate: Ask the student for an example or picture before refining the definition.

elapsed time: Ask the student for an example or picture before refining the definition.

scale: Ask the student for an example or picture before refining the definition.

line plot: Ask the student for an example or picture before refining the definition.

volume: Ask the student for an example or picture before refining the definition.

mass: Ask the student for an example or picture before refining the definition.

Material rule: The student should touch, draw, measure, or organize the mathematics before copying a procedure.

Phase 1: Launch and Number Talk

EXACT TEACHER LANGUAGE

Teacher says	"Today we are learning to collect, display, and interpret data using scaled graphs, line plots, and simple circle graphs. Before I show a method, I want to hear how you see the numbers."
Why	Positions the student as a mathematical thinker and makes prior knowledge visible.
Listen for	Useful representations, precise vocabulary, and whether the student checks reasonableness.
Then do	Acknowledge the idea neutrally, then ask for evidence rather than immediately evaluating it.

Number Talk Prompt

Teacher says	"If each picture stands for 4 votes, how many votes do 6 pictures show?"
Why	Surfaces the relationships needed in the main lesson.
Listen for	Any reasoning connected to measurement and data, not only a final answer.
Then do	Ask: 'Can you show that another way?' Record two distinct strategies when possible.

Prompt Ladder

First: What do you notice? What could you try?
Next: Can you draw, build, or organize the information?
Then: Which part of today's vocabulary fits your idea?
Last resort: Solve a smaller related case, then return to the original.

Phase 2: Explicit Teaching

MODEL, QUESTION, CONNECT

Teaching Step 1

Teacher says	"Let us collect a small data set. Tell me what I should do first and why."
Why	Keeps the student cognitively active during explicit instruction.
Listen for	A connection between the representation and the mathematical symbols.
Then do	Model only the next move. Pause and return responsibility to the student.

Teaching Step 2

Teacher says	"Let us make a scaled pictograph. Tell me what I should do first and why."
Why	Keeps the student cognitively active during explicit instruction.
Listen for	A connection between the representation and the mathematical symbols.
Then do	Model only the next move. Pause and return responsibility to the student.

Teaching Step 3

Teacher says	"Let us make a scaled bar graph. Tell me what I should do first and why."
Why	Keeps the student cognitively active during explicit instruction.
Listen for	A connection between the representation and the mathematical symbols.
Then do	Model only the next move. Pause and return responsibility to the student.

Teaching Step 4

Teacher says	"Let us make or read a simple circle graph showing category totals. Tell me what I should do first and why."
Why	Keeps the student cognitively active during explicit instruction.
Listen for	A connection between the representation and the mathematical symbols.
Then do	Model only the next move. Pause and return responsibility to the student.

Teaching Step 5

Teacher says	"Let us compare data across categories. Tell me what I should do first and why."
Why	Keeps the student cognitively active during explicit instruction.
Listen for	A connection between the representation and the mathematical symbols.
Then do	Model only the next move. Pause and return responsibility to the student.

Representation check: Before leaving the model, ask the student to point to every part of the matching equation or statement.

Phase 3: Guided and Independent Practice

RELEASE RESPONSIBILITY GRADUALLY

I Do -> We Do -> You Do

- I do: complete the packet's worked example while narrating the reason for each move.
- We do: solve the first practice task together; the student chooses the representation.
- You do: the student completes the remaining tasks while explaining selected answers.

Teacher says	"Practice task 1: A lesson starts at 9:35 and ends at 10:20. How long is it? Before solving, tell me what the task is asking and which tool might help."
Listen for	Correct interpretation before computation.
Then do	Use the prompt ladder. Avoid completing a step the student can perform.

Teacher says	"Practice task 2: Which is a reasonable mass for a watermelon: 4 grams, 4 kilograms, or 40 kilograms? Before solving, tell me what the task is asking and which tool might help."
Listen for	Correct interpretation before computation.
Then do	Use the prompt ladder. Avoid completing a step the student can perform.

Teacher says	"Practice task 3: A jug holds 2 liters. How many 250-milliliter cups fill it? Before solving, tell me what the task is asking and which tool might help."
Listen for	Correct interpretation before computation.
Then do	Use the prompt ladder. Avoid completing a step the student can perform.

Teacher says	"Practice task 4: Measurements are $3\frac{1}{4}$, $3\frac{1}{2}$, $3\frac{1}{4}$, and $3\frac{3}{4}$ inches. Which occurs most often? Before solving, tell me what the task is asking and which tool might help."
Listen for	Correct interpretation before computation.
Then do	Use the prompt ladder. Avoid completing a step the student can perform.

Teacher says	"Practice task 5: A graph scale counts by 5. A bar reaches the fourth tick above zero. What value is shown? Before solving, tell me what the task is asking and which tool might help."
Listen for	Correct interpretation before computation.
Then do	Use the prompt ladder. Avoid completing a step the student can perform.

Teacher says	"Practice task 6: Explain why units and graph scales must be written with an answer. Before solving, tell me what the task is asking and which tool might help."
Listen for	Correct interpretation before computation.
Then do	Use the prompt ladder. Avoid completing a step the student can perform.

Common Misconceptions

Watch for: Ignoring units or mixing unlike units.

Watch for: Reading a graph without checking its scale or key.

Watch for: Treating clock minutes like base-ten digits.

Differentiation

Secure: Remove routine repetition and ask for a second method, proof, or generalization.

Developing: Keep the visual model visible and reduce the number size while preserving the same structure.

Fragile: Return to a concrete example, use one question at a time, and delay independent symbols.

Phase 4: Challenge, Assessment, and Feedback

PROTECT PRODUCTIVE STRUGGLE

Teacher says	"Challenge: Create a graph where the scale is 3. Write a question that requires multiplication and subtraction. You may draw, make a table, test a simpler case, or work backward. I will give you thinking time before offering a hint."
Why	Builds persistence and strategy selection rather than rapid answer production.
Listen for	An organized plan, revision after a failed attempt, and a justification.
Then do	Praise a specific mathematical action: organization, checking, representation, or explanation.

Exit Check

Teacher says	"Complete this independently: Why must you read the key before answering questions about a scaled graph?"
Why	Provides evidence of the central lesson goal without teacher prompting.
Then do	After completion, ask one clarifying question only if the written reasoning is ambiguous.

Evidence Guide

Secure	Correct and independently justified	Proceed; add a variation or ask the student to teach the idea.
Developing	Correct with model or prompt	Proceed with the model available and begin next lesson with retrieval.
Fragile	Incorrect structure or unexplained answer	Reteach with smaller quantities and a concrete representation.

Answer and Observation Guide

USE AFTER INDEPENDENT WORK

Expected Mathematical Evidence

- Number talk: a defensible response to 'If each picture stands for 4 votes, how many votes do 6 pictures show?' with a model or explanation.
- Main lesson: evidence that the student can collect, display, and interpret data using scaled graphs, line plots, and simple circle graphs.
- Practice: accurate work with units, labels, and a reasonableness check where appropriate.
- Challenge: an organized attempt at 'Create a graph where the scale is 3. Write a question that requires multiplication and subtraction.' and an explanation of the chosen strategy.
- Exit: an independent response to 'Why must you read the key before answering questions about a scaled graph?'

Teacher Verification Routine

- Rework every numerical answer before marking it.
- Accept a different method when the reasoning is valid.
- For open problems, verify that every condition is satisfied.
- For measurement, data, and geometry, require correct units and labels.
- For explanations, distinguish a correct statement from a demonstrated reason.

Assessment Answer Notes

1. 45 minutes.
2. 4 kilograms.
3. 8 cups.
4. 3 1/4 inches.
5. 20.
6. Explanation should state that units and scales determine the quantity represented.

Observation Record

A strong strategy the student used:

A misconception to revisit:

Vocabulary used precisely:

Material or representation that helped:

Plan for the next lesson:
